

Charge-Transfer Complexes

Through-Space Charge Transfer Complexes Based on Terpyridine Bi-Functionalization for Efficient Solar-Driven H₂O₂ Production

Manman Dai, Ningxu Han, Xiang Li, Junjuan Shi, He Chen, Zhihong Chen,* and Ming Wang*

Abstract: To achieve precise spatial control of donor–acceptor (D–A) stacking and optimize through-space charge transfer (TSCT) for efficient photocatalysis, we develop a novel terpyridine bi-functionalization strategy, which utilizes the dissymmetrical 6- and 4'-functionalization of 2,2':6',2''-terpyridine (tpy) scaffold with triphenyltriazine (acceptor) and triphenylamine (donor) units. Using this strategy, we designed and synthesized two ligands, **L1** and **L2**, differing by a single phenylene unit in length. Upon coordination with Zn(II), the D and A units in the resulting complexes **S1** and **S2** are precisely edited at the molecular level, giving rise to distinct stacking geometries: slip-stacked **S1** and co-facially **S2**. **S2** significantly enhances TSCT, resulting in improved visible-light absorption, prolonged charge carrier lifetime, superior charge separation efficiency and achieves a photocatalytic efficiency (2063 $\mu\text{mol g}^{-1} \text{h}^{-1}$) for H₂O₂ production that is twice that of **S1** without the need for sacrificial agents. This work demonstrates the unique advantages of terpyridine bi-functionalization strategy in constructing photocatalytic materials with tunable TSCT, providing new insights for the design and functional development of supramolecular systems.

Introduction

Hydrogen peroxide (H₂O₂) is a dual-functional chemical of critical importance, serving both as a versatile green oxidant and a promising high-energy-density liquid fuel.^[1–5]

Its broad utility spans chemical manufacturing, textiles, and sterilization, while its emergence as an environmentally benign alternative to highly toxic hydrazine-based propellants (e.g., unsymmetrical dimethylhydrazine) in aerospace applications underscores its strategic value.^[6–8] Industrially, H₂O₂ production remains dominated by the energy-intensive, multi-step anthraquinone (AQ) process, which generates significant waste streams.^[9] These limitations have intensified efforts toward sustainable, low-energy synthetic routes.^[10,11] Among them, solar-driven photocatalysis—particularly systems utilizing pure water and atmospheric O₂—has emerged as a highly attractive approach. However, existing photocatalysts suffer from inherent inefficiencies, including poor visible-light absorption, rapid charge recombination, reliance on sacrificial agent, and noble metals, low H₂O₂ selectivity, and insufficient activity.^[12–15] Consequently, the development of high-performance catalysts capable of efficient, sacrificial-agent-free, and noble-metal-free H₂O₂ generation under ambient conditions remains an urgent challenge.

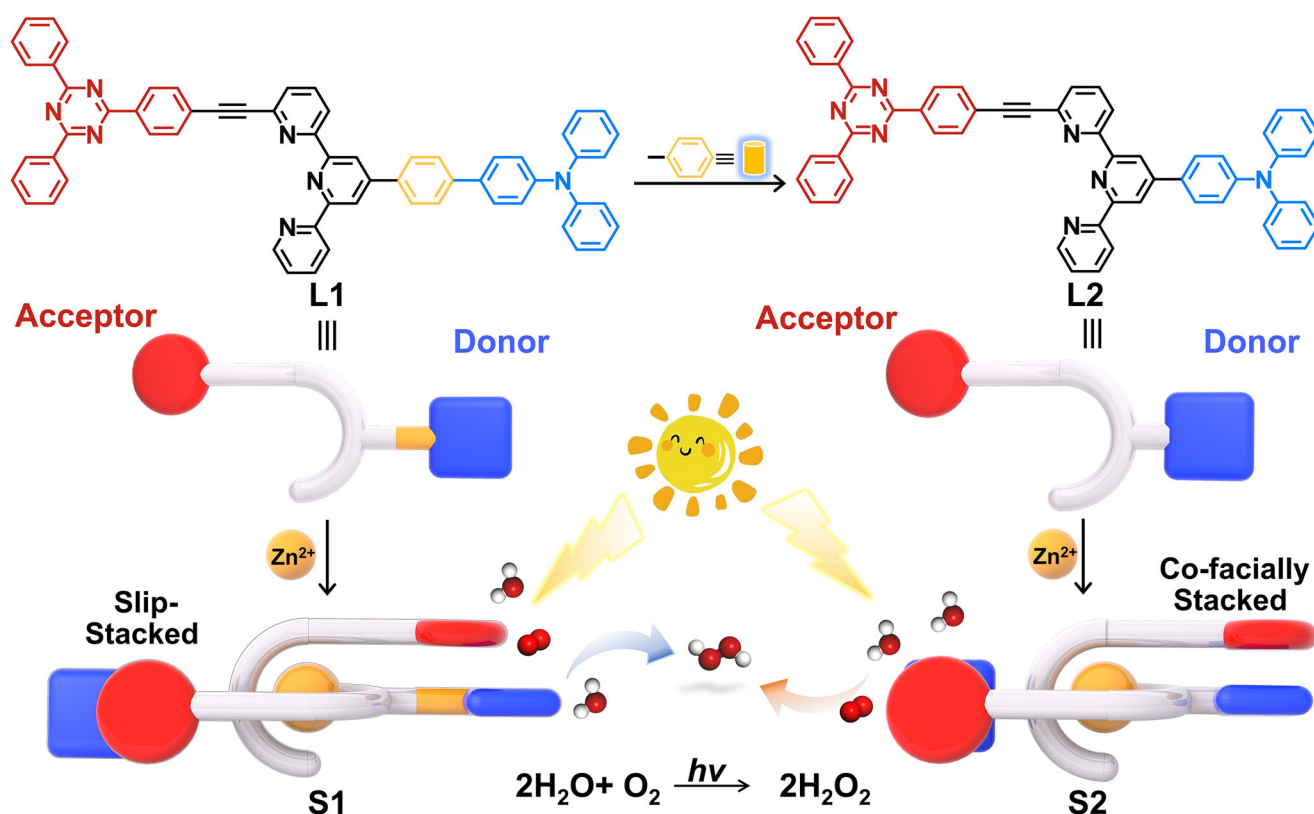
Donor–acceptor (D–A) featured molecular systems offer a powerful platform to overcome the limitations of conventional photocatalysts.^[16–19] Through the deliberate modulation of energy-level offsets, these systems facilitate directional charge separation and suppress recombination,^[20,21] while extending visible-light response and optimizing redox potentials.^[22,23] As a result, D–A systems significantly enhance photocatalytic efficiency. Among the various charge-transfer mechanisms, through-space charge transfer (TSCT) which occurs between spatially adjacent D and A units within a donor–spacer–acceptor (D–S–A) framework offers distinct advantages over conventional through-bond charge transfer (TBCT).^[24–26] The enforced proximity between D and A via a molecular spacer enhances electronic coupling, enabling rapid charge separation with minimal energy loss.^[27–29] Importantly, the inherent modularity of TSCT architectures affords unparalleled flexibility in tuning intramolecular spatial organization—a key asset for photocatalytic design. Key strategies to optimize D–A interactions in TSCT systems include: i) geometric confinement (pre-organizing D–A spatial arrangements via rigid scaffolds);^[30–32] ii) linker mediation (modulating electronic communication through spacer engineering);^[25,33–35] iii) aggregative coupling (exploiting supramolecular interactions, e.g., π – π stacking/ H-bonding).^[26,27,36–38] Despite the availability of these strategies, achieving atomically precise control over D–A distance and orientation to maximize TSCT remains a formidable challenge—a bottleneck limiting the rational design of advanced photocatalysts.

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Scheme 1. Design and self-assembly of ligands and complexes.

Coordination-driven self-assembly is a bottom-up approach for constructing well-defined metallo-supramolecules with precisely controlled shapes and sizes. Compared with covalent bonds, coordination bonds, as a type of non-covalent interaction, can enable reactions to proceed efficiently under thermodynamic control.^[39] Compared with covalent interactions, coordination-driven self-assembly offers operational simplicity and high reaction yields.^[40] Moreover, among various non-covalent interactions, coordination bonds based on metal–ligand interactions have high directionality and predictable features, due to the defined geometry of metal–ligand coordination. They also provide greater stability than other types of non-covalent bonds.^[41] Therefore, coordination-driven self-assembly is an effective approach for achieving precise intramolecular chromophore stacking,^[42,43] which facilitates for the realization of TSCT and thereby ultimately contributes to the development of highly efficient photocatalytic materials for promising applications like H_2O_2 production.^[44] In this work, we utilize coordination-driven self-assembly approach to enforce geometrically confined, co-facial D–A stacking through subtle linker editing. Our approach leverages a programmable 2,2':6',2''-terpyridine (tpy) scaffold, asymmetrically functionalized at the 6- and 4'-positions with a triphenylamine (TPA) donor and a triphenyltriazine (TTz) acceptor. By synthesizing two ligands (**L1** and **L2**) with systematically varied linker lengths between TPA and the tpy core, and coordinating them with non-noble Zn(II), we constructed mononuclear complexes **S1** and **S2**. These

complexes feature well-defined spatial configurations and controlled D–A orientations, giving rise to distinct TSCT characteristics. Single-crystal X-ray analyses revealed that a reduction in linker length by a single phenylene unit drives a structural metamorphosis from a slip-stacked architecture in **S1** (3.8 Å) to a co-facially stacked motif in **S2** (3.5 Å), thereby significantly enhancing TSCT. This subtle geometric optimization translated directly into a remarkable photocatalytic enhancement: under identical sacrificial-agent-free conditions (using only air and water, 0.1 g L^{−1} catalyst loading, visible light), **S2** achieved an H_2O_2 production rate of 2063 $\mu\text{mol g}^{-1} \text{h}^{-1}$, nearly twice that of **S1** (1099 $\mu\text{mol g}^{-1} \text{h}^{-1}$), highlighting the profound impact of minimal molecular editing on catalytic function. Furthermore, **S2** exhibited excellent stability, maintaining its structural and crystallographic integrity over five consecutive cycles. This work not only delivers a robust platform for sustainable H_2O_2 production but also establishes coordination-driven spatial engineering as a general paradigm for elucidating structure–activity relationships in D–A photocatalysts, paving the way for the rational design of advanced TSCT materials.

Results and Discussion

Synthesis and characterization of ligands and complexes. As shown in Scheme 1, ligands **L1** and **L2** incorporate TTz unit at the 6-position of the tpy core, while differing at the 4'-position: **L1** features a phenyl-extended TPA derivative,

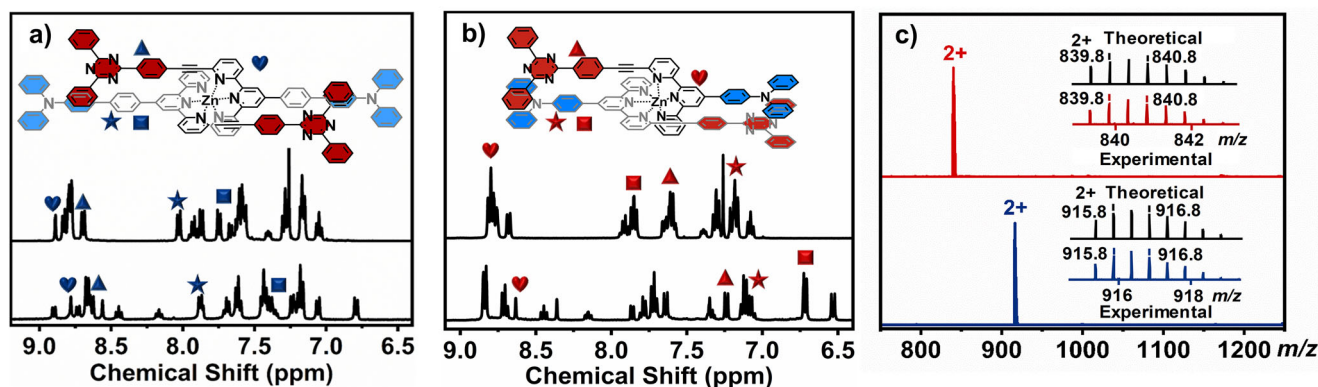


Figure 1. ^1H NMR spectra (400 MHz, 298 K, aromatic region) of a) **L1** (top) in CDCl_3 and **S1** (bottom) in CD_3CN and b) **L2** (top) in CDCl_3 and **S2** (bottom) in CD_3CN . c) ESI-MS spectra of **S1** (blue line) and **S2** (red line).

whereas **L2** bears a non-extended TPA unit. The synthetic routes of these two ligands were shown in Scheme S1. This structural variation results in long-chain (**L1**) and short-chain (**L2**) ligand architectures, respectively. **S1** and **S2** were obtained by assembling the ligands with $\text{Zn}(\text{II})$ in 2:1 molar ratio in $\text{CHCl}_3/\text{CH}_3\text{OH}$ (1:2, v/v) solvent mixture at 50°C for 4 h. Precipitation with excess NH_4PF_6 in methanol afforded orange–yellow (**S1**) and orange–red (**S2**) solids in yields exceeding 95%. All compounds were thoroughly characterized by ^1H NMR (Nuclear Magnetic Resonance Spectroscopy), ^{13}C NMR, 2D COSY (Two Dimensional Correlation Spectroscopy), ESI-MS (Electrospray Ionization Mass Spectrometry) and TWIM-MS (Traveling-Wave Ion Mobility Mass Spectrometry) (Figures S1–S40). The ^1H NMR spectra of these two ligands exhibit three distinct pyridyl proton signals, which is consistent with their dissymmetrical structures (Figure 1a,b). Upon coordination, significant upfield shifts were observed for key protons: the signal of $\text{tpy-H}^{3'}$ in **S1** shifted from 8.89 ppm (**L1**) to 8.78 ppm, while in **S2**, it moved from 8.79 ppm (**L2**) to 8.63 ppm. Additionally, the aromatic protons of $\text{Ph-H}^{i,j}$ displayed marked upfield shifts ($\Delta\delta = 0.76$ – 1.12 ppm). These shifts are attributed to the interactive shielding effect between the phenylene and pyridyl moieties, supporting the formation of a head-to-tail coordination motif. ESI-MS confirmed the mononuclear composition: deconvoluted spectra yielded molecular weights of 2120.5 and 1968.5 Da (Figure 1c). Experimental isotope patterns matched closely with simulated isotope pattern (Figure S1). TWIM-MS showed narrow drift time distributions for all charge states, indicative of discrete, rigid assemblies devoid of conformational or isomeric heterogeneity (Figure S2).^[45] Thermogravimetric analysis (TGA) demonstrated the excellent thermal stability for both complexes up to 200°C under a nitrogen atmosphere (Figure S3) (Figure 2).

Single-crystal structures of these two complexes were obtained via slow vapor diffusion of ethyl acetate into acetonitrile solutions of two complexes over 4 weeks. X-ray crystallography analysis confirmed a head-to-tail coordination geometry (Figures 2a–b and S41–S42), with both complexes adopting a “single-winged butterfly” architecture: the tpy core forms the butterfly body, while the TTz and TPA

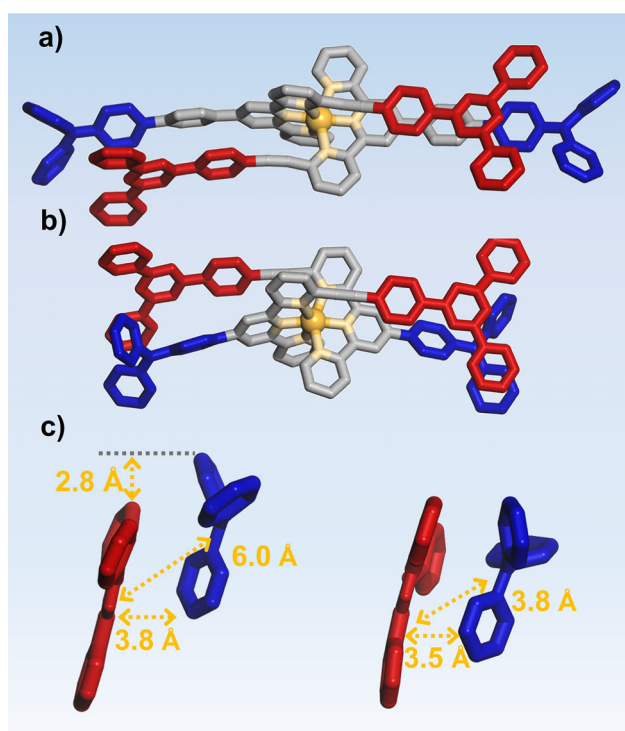


Figure 2. Single-crystal structures of a) **S1** and b) **S2**, respectively. c) Side views of (left) **S1** and (right) **S2** only display the donor-acceptor structure on one side of the complexes. Hydrogen atoms, counterions, and solvents are omitted for clarity.

units constitute the wings. In **S1**, the extended phenyl linker facilitates parallel-displaced stacking between the D and A units, which features an interlayer distance of 3.8 Å and a distance of 6.0 Å from the N atom in TPA to the centroid of TTz. Conversely, **S2** results in face-to-face stacking, which has an interlayer distance of 3.5 Å and a distance of 3.8 Å from the N atom in TPA to the centroid of TTz. Notably, the bulky PF_6^- counterions and the butterfly motif contribute to weak π – π interaction (Figures S43–S46). Solid-state analysis revealed weak intermolecular π – π interactions in **S1** but no significant interactions in **S2**, suggesting minimal electronic perturbation from the crystalline lattice. These

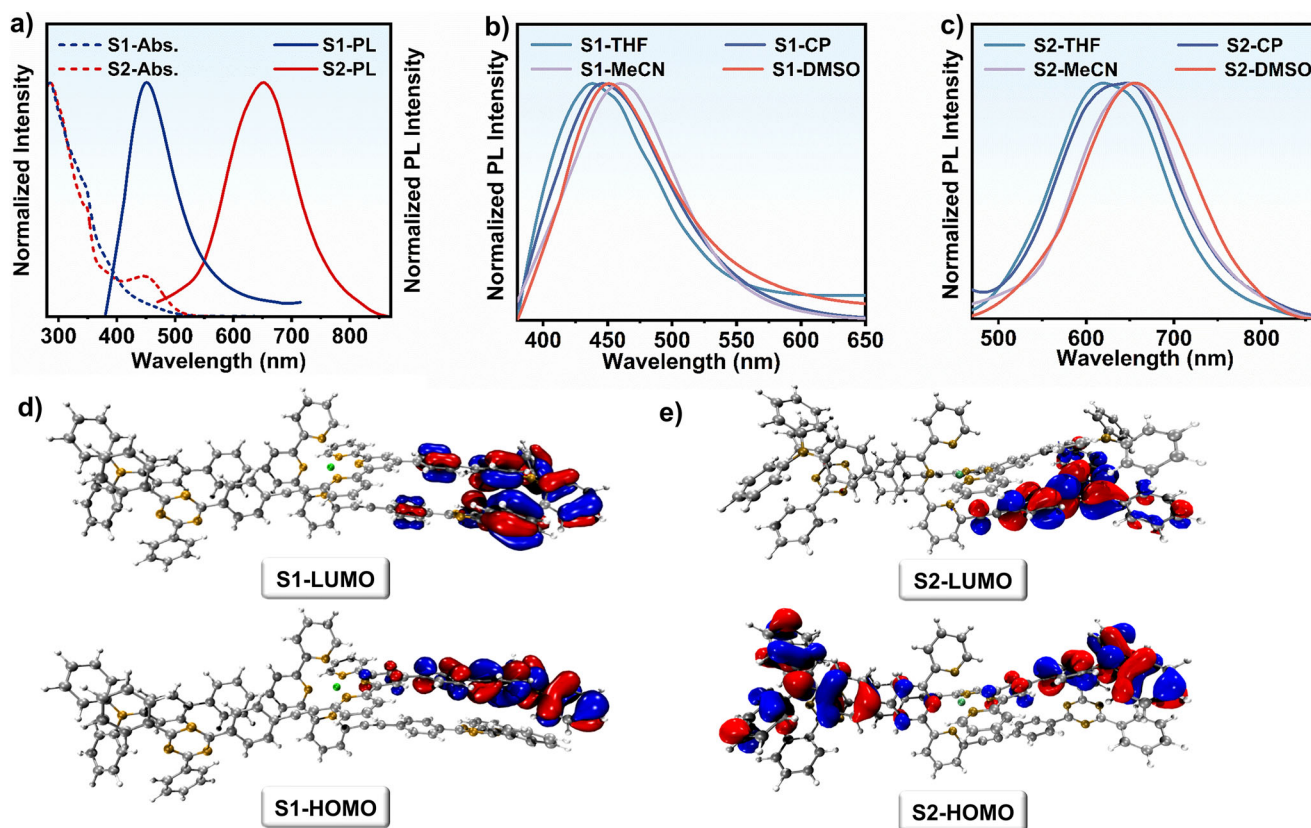


Figure 3. a) UV-vis absorption and fluorescence emission spectra of **S1** and **S2** in DMSO ($c = 10.0 \mu\text{M}$). The photoluminescence spectra in different solvents at 298 K of b) **S1** and c) **S2**. DMSO (dimethyl sulfoxide); CH₃CN (acetonitrile); CP (acetone); and THF (tetrahydrofuran). Frontier molecular orbitals involved in the electronic transitions for d) **S1** and e) **S2** obtained by DFT calculations.

structural distinctions guided subsequent investigations into their photophysical behavior.^[46]

Absorption and Fluorescence Properties

The photophysical properties of ligands **L1**, **L2** and complexes **S1**, **S2** in solution state were systematically investigated (Figures S47–S49 and Tables S3 and S4). As shown in Figure S47, the UV-vis absorption spectra of complexes **S1** and **S2** exhibit significant red-shift relative to their corresponding ligands, with new bands emerging in the 400–500 nm range. This indicates the emergence of intramolecular charge transfer (ICT) transitions upon coordination.^[47] Fluorescence emission spectra (Figure 3a) show that the short-chain complex **S2** displays a substantially larger Stokes shift (nearly 200 nm) than the long-chain analogue **S1**, with its emission extending into the near-infrared (NIR) region. A red shift of the emission band was observed in the order of **S1** (450 nm) and **S2** (657 nm), possibly the more pronounced red-shift of **S2** can be attributed to its optimized coplanar conformation between the TPA and TTz units, facilitating stronger ICT.^[48] In contrast to **S1**, which shows emission similar to ligand **L1**, **S2** displays broader and more red-shifted emission relative to ligand **L2**, demonstrating inefficient ICT for **S1** and significantly enhanced ICT for **S2** (Figure 3b,c).^[49] To

elucidate the charge-transfer mechanism, density functional theory (DFT) calculations were performed (Figure 3d,e). The slip-stacked **S1** exhibits incomplete spatial separation between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO). In contrast, the co-facially stacked **S2** shows ideal separation, with the HOMO located on electron-donating TPA and the LUMO located on electron-withdrawing TTz. This well-separated electronic structure precisely corresponds to the characteristic signature of TSCT process. Collectively, **S2** manifests superior TSCT characteristics, with its photophysical properties directly governed by intramolecular geometry. Thus, intramolecular geometry directly dictates TSCT capability, with **S2** exhibiting larger Stokes shift, ideal spatial separation of orbitals through enforced co-facial alignment.^[50]

Electrochemical Testing

The optical and electronic properties of both complexes were thoroughly examined by ultraviolet-visible diffuse reflection spectroscopy (UV-vis DRS). As shown in Figures 4a and S50, both complexes **S1** and **S2** demonstrated significant light absorption across a broad spectrum of visible light, highlighting their efficient light-harvesting capabilities compared to

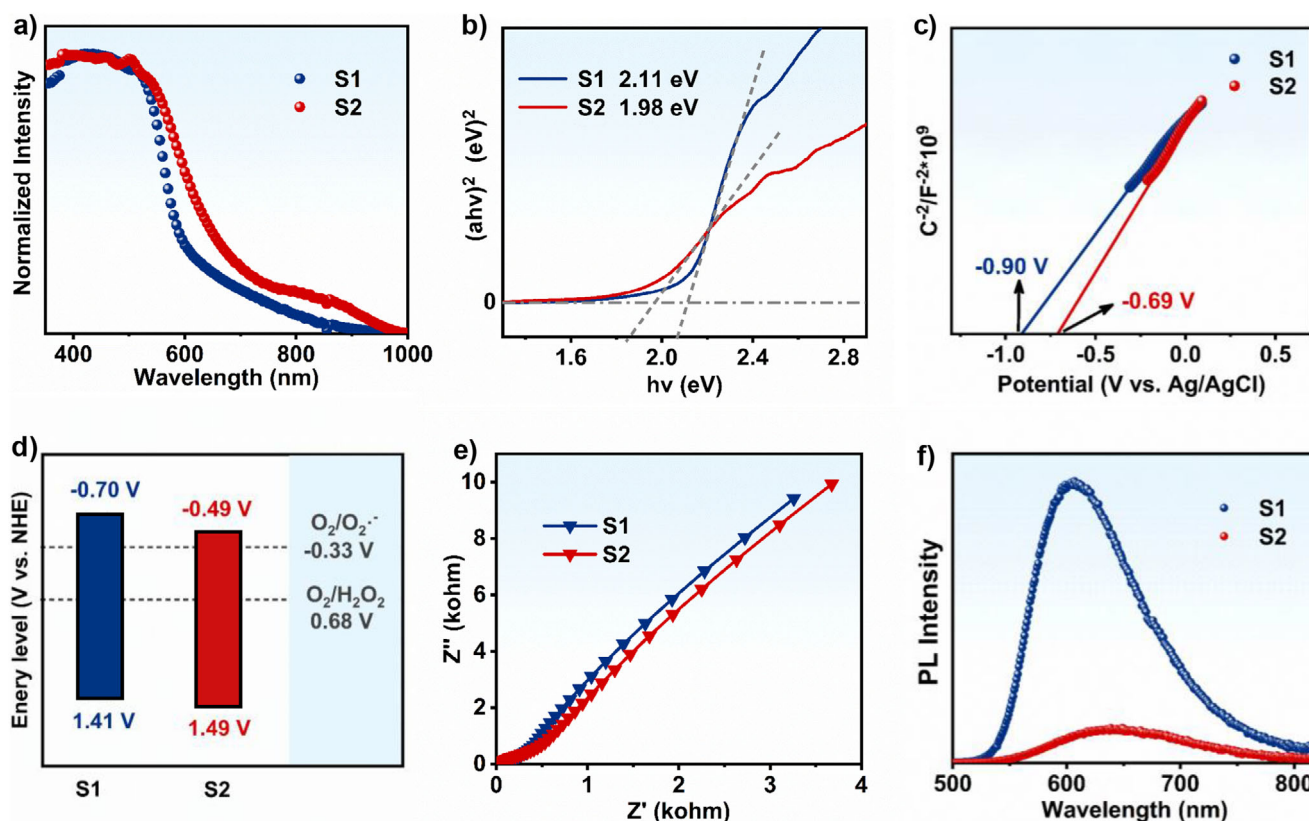


Figure 4. a) UV-vis diffuse reflectance spectra and b) the Tauc plot curves of **S1** and **S2**. c) Mott-Schottky plots of **S1** and **S2** at frequency of 1000 Hz. d) Energy level diagrams of **S1** and **S2**. e) Nyquist plots and f) PL spectra of **S1** and **S2**.

the narrow absorption range (200–500 nm) of the ligands. Tauc plots derived from Kubelka–Munk transformations (Figure 4b) yield optical bandgap (E_g) values of **S1** and **S2** are 2.11 and 1.98 eV, which represent their semiconductor characteristics.^[51] **S1** and **S2** were n-type semiconductors, as indicated by the positive slopes of the Mott–Schottky plots at the isoelectric point in Figures 4c and S51, with flat-band potentials of -0.90 and -0.69 V (versus Ag/AgCl), respectively. For n-type semiconductors, the conduction band (CB) potential is normally 0.1–0.2 V higher than the flat-band potential.^[52,53] The CB potentials of these complexes were estimated to be -0.70 and -0.49 V (versus NHE), respectively. These values were lower than the reduction potentials for both the direct $2e^-$ oxygen reduction reaction (ORR) (+0.68 V versus NHE) and the indirect $2e^-$ ORR (-0.33 V versus NHE), suggesting that all the complexes possess thermodynamically favorable energy levels to facilitate H_2O_2 production via ORR. In addition, the valence band (VB) positions of **S1** and **S2** are calculated as +1.48 and +1.49 V (versus NHE) respectively, which were lower than the potentials required for the direct $2e^-$ water oxidation reaction (WOR) (+1.76 V versus NHE). Therefore, **S1** and **S2** were thermodynamically favorable for H_2O_2 photogeneration via ORR pathway in pure water; **S1** and **S2** may be less efficient in driving the $2e^-$ WOR pathway due to its lower VB potential, thereby limiting its efficiency in H_2O_2 production through water oxidation.

Photoelectrochemical profiling further highlights the superiority of **S2**: transient photocurrent density measurements (Figure S52) show that **S2** achieves the highest response, indicating optimal interfacial electron transfer; electrochemical impedance spectroscopy (EIS) reveals smallest Nyquist arc radius for **S2** (Figure 4e), reflecting minimal charge-transfer resistance; steady-state PL spectra confirm significant fluorescence quenching in **S2** (Figure 4f), directly evidencing suppressed carrier recombination; time-resolved PL decay curves give longer average lifetimes for **S2** (13.63 ns versus 5.70 ns for **S1**), quantitatively supporting enhanced charge separation kinetics (Figure S53 and Table S5). These multifaceted results validate **S2**'s superior photocatalytic potential arising from optimized charge separation and transport properties.

Photocatalytic H_2O_2 Production

The photocatalytic performance of complexes **S1** and **S2** for H_2O_2 production in a sacrificial agent-free aqueous system under Xe lamp irradiation is presented in Figure 5a. The H_2O_2 yield of **S2** reaches $2063 \mu\text{mol g}^{-1} \text{h}^{-1}$, which is almost two times higher than that of the slipped D–A stacked **S1** ($1099 \mu\text{mol g}^{-1} \text{h}^{-1}$). Notably, this performance outperforms most reported photocatalysts, even those operating in sacrificial agent-containing systems (Figure S55 and Table

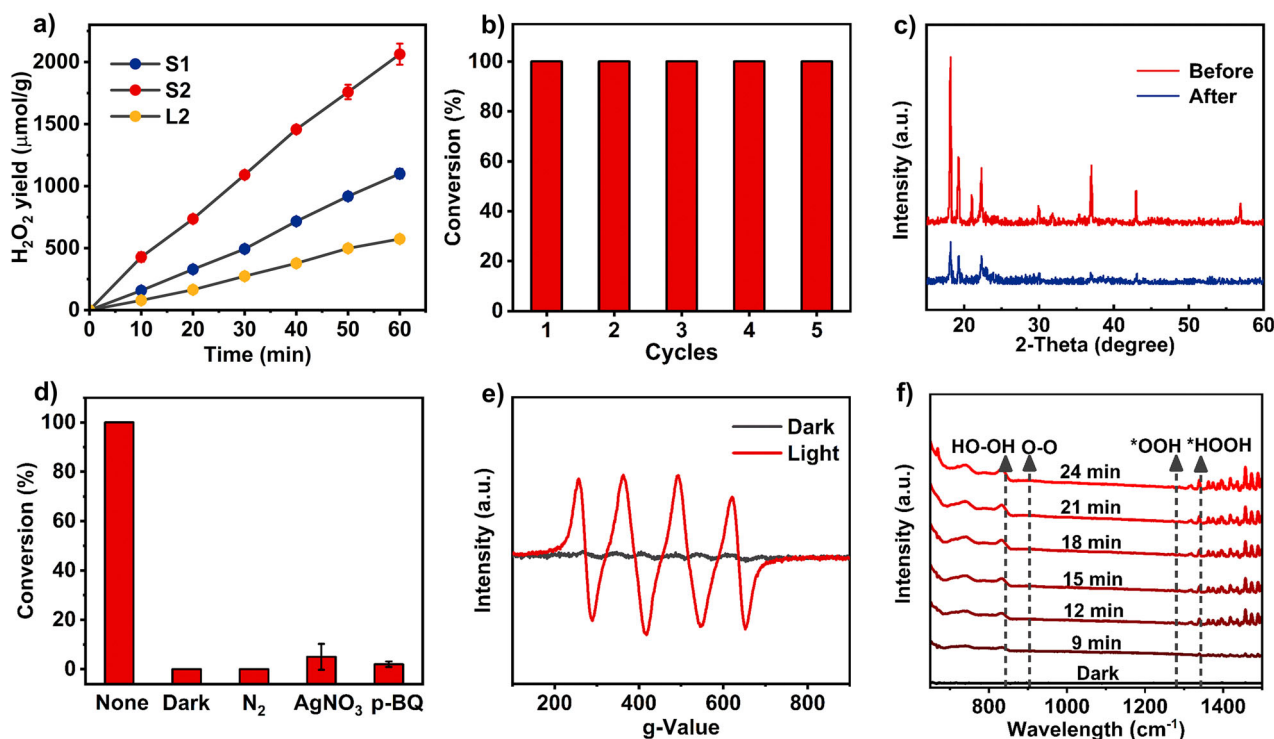


Figure 5. a) The performances of complexes **S1**, **S2**, and ligand **L2** for H_2O_2 photoproduction in water in the air atmospheres. b) H_2O_2 photosynthesis performance of **S2** running for five consecutive cycles. c) PXRD spectra of **S2** before and after photocatalysis. d) H_2O_2 production of **S2** with different scavengers. EPR trapping experiments in the presence of BMPO e) before (gray line) and after (red line) light irradiation of **S2**. f) In situ DRIFTS spectra of **S2** recorded during photocatalytic H_2O_2 production.

S6). Stability assessments reveal that **S2** retains 98% of its initial activity after five consecutive cycles (Figure 5b). Post-cycling characterization by powder X-ray diffraction (PXRD) shows no shift in main diffraction peaks (Figure 5c), confirming the excellent chemical and structural stability of **S2**. Moreover, the ^1H NMR, UV-vis DRS and UV-vis absorption measurements suggest the stability of catalyst **S2** after five consecutive cycles, with negligible change in its chemical characteristics, confirming its excellent stability and well-preserved structure (Figures S56 and S57).

Photocatalytic H_2O_2 generation is known to proceed via two potential pathways: ORR and WOR.^[54] To determine the dominant pathway for **S2**, a series of control experiments were conducted. As shown in Figure 5d, switching the reaction atmosphere from air to N_2 or dark conditions caused a dramatic drop in H_2O_2 generation rate. These results indicated that the H_2O_2 was dominantly produced from the 2e^- ORR route using O_2 as raw materials, instead of WOR with $\cdot\text{OH}$ as key intermediates.^[55] Furthermore, the active intermediate was verified by introducing different scavengers, i.e., benzoquinone (p-BQ, $\cdot\text{O}_2^-$ scavenger), AgNO_3 (e^- scavenger) (Figure 5d). AgNO_3 strongly suppressed the H_2O_2 production, suggesting that photogenerated electrons are involved in the ORR process.^[56] Also, no H_2O_2 was produced in the presence of p-BQ, which means that $\cdot\text{O}_2^-$ participates in the formation of H_2O_2 . Electron paramagnetic resonance (EPR) spectroscopy provided direct evidence for reactive oxygen species (ROS) involvement. Using 5-(*tert*-Butoxycarbonyl)-5-methyl-1-pyrroline-N-oxide (BMPO) as

a spin trap, a distinct signal of superoxide anion radicals ($\cdot\text{O}_2^-$) was detected (Figure 5e). In contrast, no observable $\cdot\text{OH}$ signal was found in the 5,5-dimethyl-1-pyrroline-N-oxide (DMPO)-trapped $\cdot\text{OH}$ EPR spectrum (Figure S58), further excluding $\cdot\text{OH}$ as a significant reactive species. These experiments demonstrated that the produced H_2O_2 over **S2** was mainly through a 2e^- ORR progress.

Photocatalytic mechanism of **S2**

Based on the aforementioned mechanistic investigations, as shown in Figure 6, the reaction mechanism of **S2**-catalyzed H_2O_2 generation under visible light. The process primarily follows the indirect 2e^- ORR pathway: O_2 is first adsorbed on the **S2** surface and reduced to $\cdot\text{O}_2^-$ via single-electron transfer ($\text{O}_2 + \text{e}^- \rightarrow \cdot\text{O}_2^-$); subsequently, $\cdot\text{O}_2^-$ undergoes further electron and proton acquisition to generate H_2O_2 ($\cdot\text{O}_2^- + 2\text{H}^+ + \text{e}^- \rightarrow \text{H}_2\text{O}_2$). The exceptional performance of **S2** is attributed to its co-facial D-A stacking architecture, which optimizes O_2 adsorption (Figure S59) and electron transfer efficiency—key factors for efficient solar-driven H_2O_2 synthesis.

Conclusion

In this work, we utilize the *tpy* bi-functionalization strategy to design and synthesize two similar ligands **L1** and **L2**, which

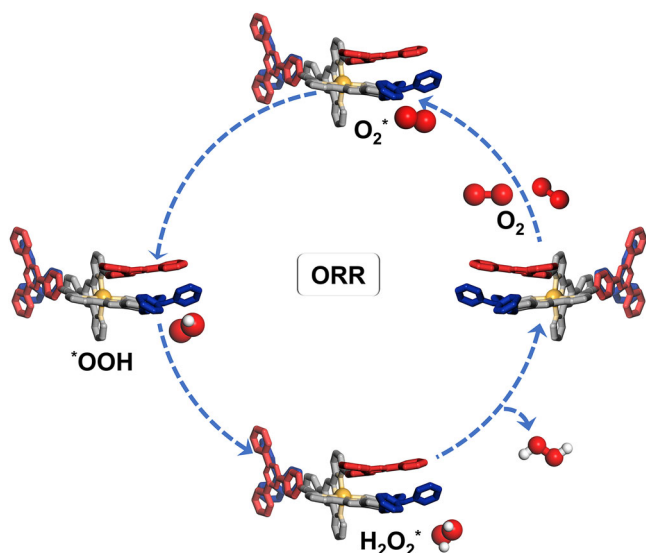


Figure 6. Schematic diagram of photocatalytic H_2O_2 production through the ORR pathways over **S2**.

have identical donor (TPA) and acceptor (TTz), differing by only single phenylene. By coordinating them with non-noble Zn(II), two distinct mononuclear complexes, **S1** and **S2**, were successfully obtained. The subtle difference in the ligands leads to significant variations in the arrangement of the D–A units within the assembled structures, shifting the D–A stack from slip (**S1**, 3.8 Å) to co-facial (**S2**, 3.5 Å) verified by single-crystal X-ray diffraction. Due to the significant intramolecular TSCT, these complexes are utilized for photocatalytic H_2O_2 production. These complexes exhibit excellent photocatalytic properties in the absence of sacrificial agents with a catalyst loading of 0.1 g L^{-1} and excellent stability. Interestingly, although these two complexes have similar structure, the **S2** with co-facial stacking shows a substantially larger Stokes shift (nearly 200 nm) and improved visible-light absorption, prolonged charge carrier lifetime, and superior charge separation efficiency—all resulting from efficient TSCT. The photocatalytic efficiency for H_2O_2 production reaches 2063 $\mu\text{mol g}^{-1} \text{h}^{-1}$ under air and water conditions, which is twice that of **S1**. This work provides a new and universally applicable design approach, which offers valuable insights for molecular assembly, complex design and the development of functional materials.

Supporting Information

The authors have cited additional references within the Supporting Information.^[57–76] Synthetic details and characterization of ligands and complexes including NMR, ESI-MS, TWIM-MS, single crystals, and additional performance tests.

Accession Codes: CCDC 2481579 and 2481624 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Single crystal • Supramolecular architectures • Terpyridine • Through-space charge transfer

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